

Visualization Design:

Visual Encodings and Scale, Day 2

Sarah Groves

Reading

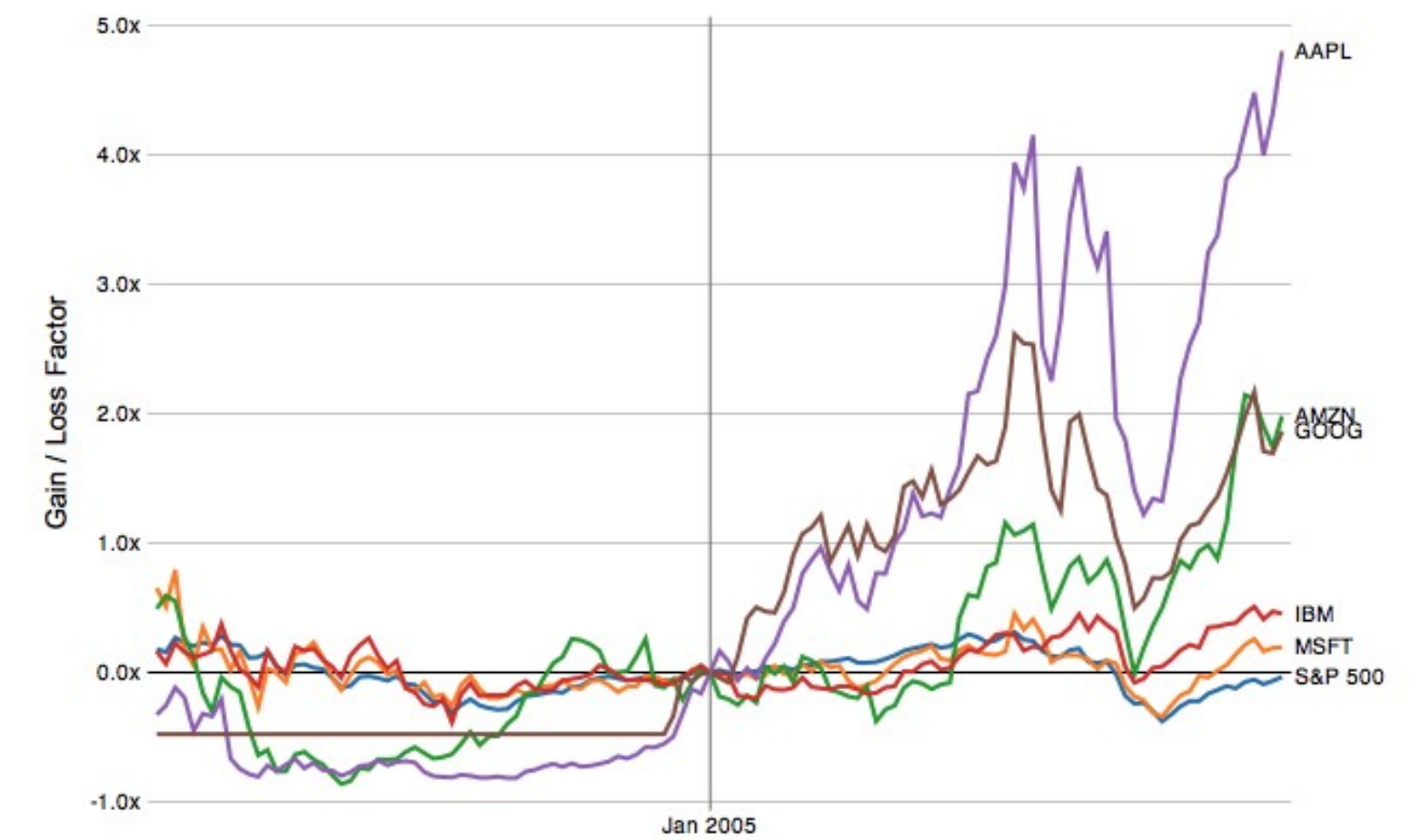
A Tour through the Visualization Zoo by Jeffrey Heer, Michael Bostock, and Vadim Ogievetsky

Overview of various visualization types

Some are common, some are “exotic”, thus it is a tour through “the zoo”

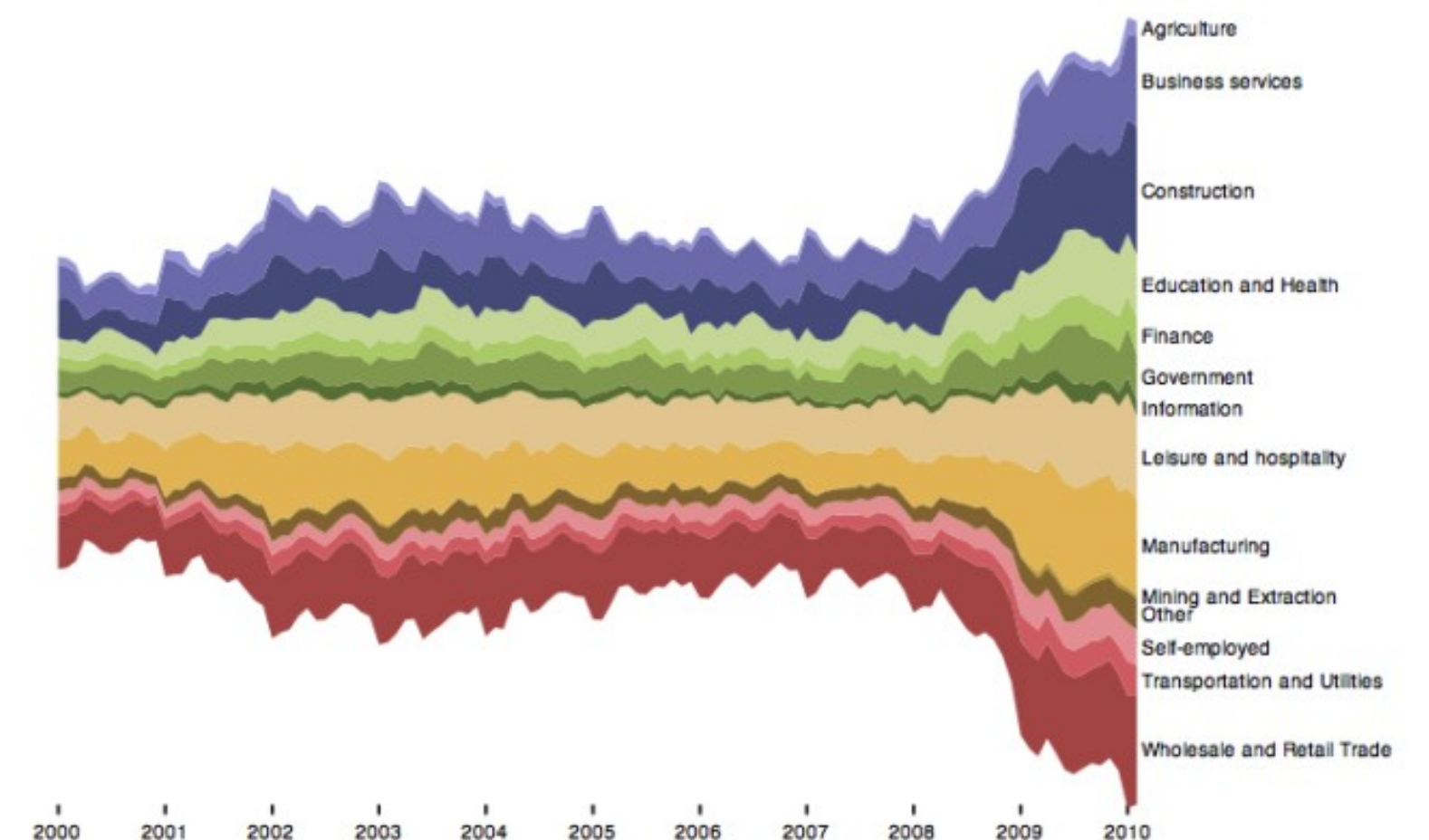
Please read by next Wednesday!
Discussion post on Canvas

Index Charts



With some forms of time-series data, raw values are less important than relative changes. Consider investors, who are more interested in a stock's growth rate than its specific price. Multiple stocks may have dramatically different baseline prices, but be meaningfully compared when normalized. An *index chart* is an interactive line chart that shows percentage changes for a collection of time-series based on a selected index point. For example, the image above shows the percentage change of selected stock prices if purchased in January 2005: one can see the rocky rise enjoyed by those who invested in Amazon, Apple, or Google at that time.

Stacked Graphs



The Big Picture



Physical Data Types

int, float, string

Conceptual Data Types

temperature, location

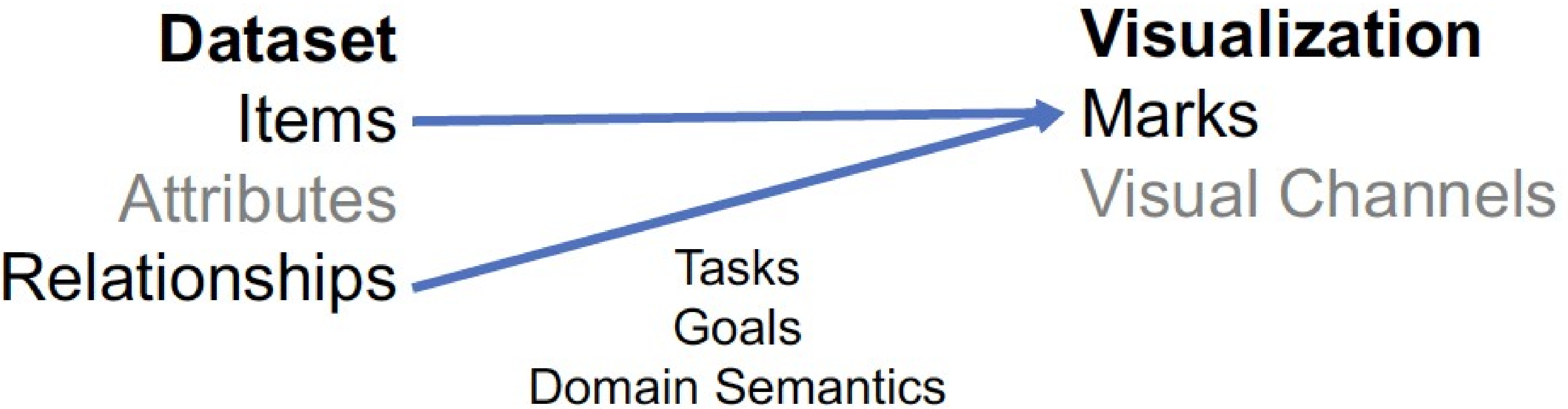
Visual Channels

x, y, color, opacity

Graphical Marks

rect, line, point, area

Goal of Visual Encoding



Deconstructing charts

Walk around the room and look at the charts.

Write out **one mapping**:

- A mapping between **data items and marks**
- A mapping between **data attributes and visual channels**

If all mappings are already written out by other students, write your answer to the following about a particular mapping: **is it a good one?**

Marks as Items

→ Points



→ Lines

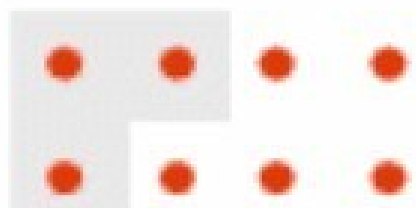


→ Areas



Marks as Relationships

→ Containment



→ Connection



Attributes as Channels

→ Position

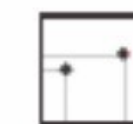
→ Horizontal



→ Vertical



→ Both



→ Shape



→ Color (Hue,Saturation,Value)



→ Tilt

/Angle/Orientation



→ Size

→ Length



→ Area



→ Volume



A visual encoding defines

- A mapping between data items and marks
- A mapping between data attributes and visual channels
- Scale: type, domain, range, aggregator

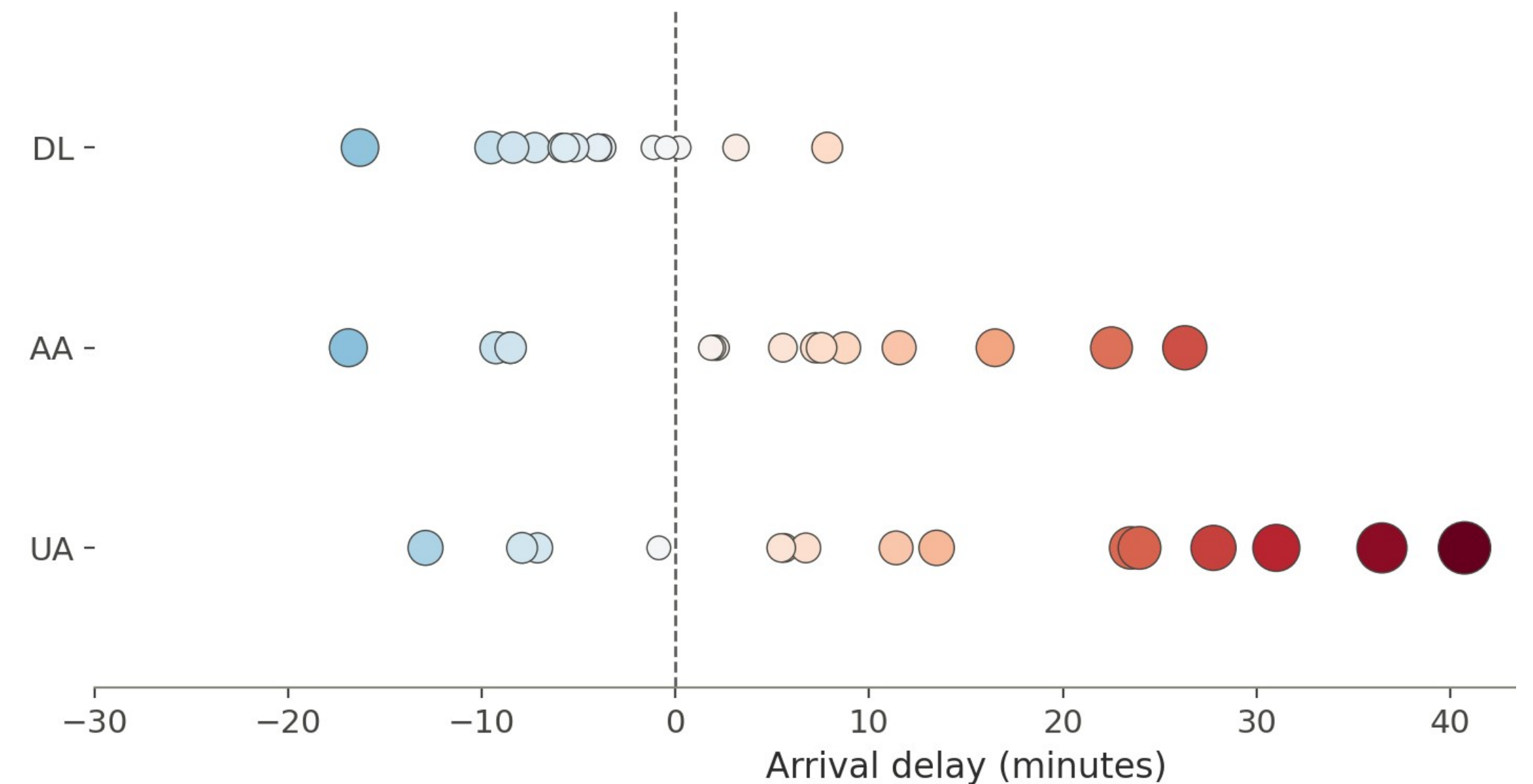
What is a scale?

A function specifying **how** abstract data values are mapped to visual properties

Dimensions of a scale

- **Domain:** the data values to be mapped

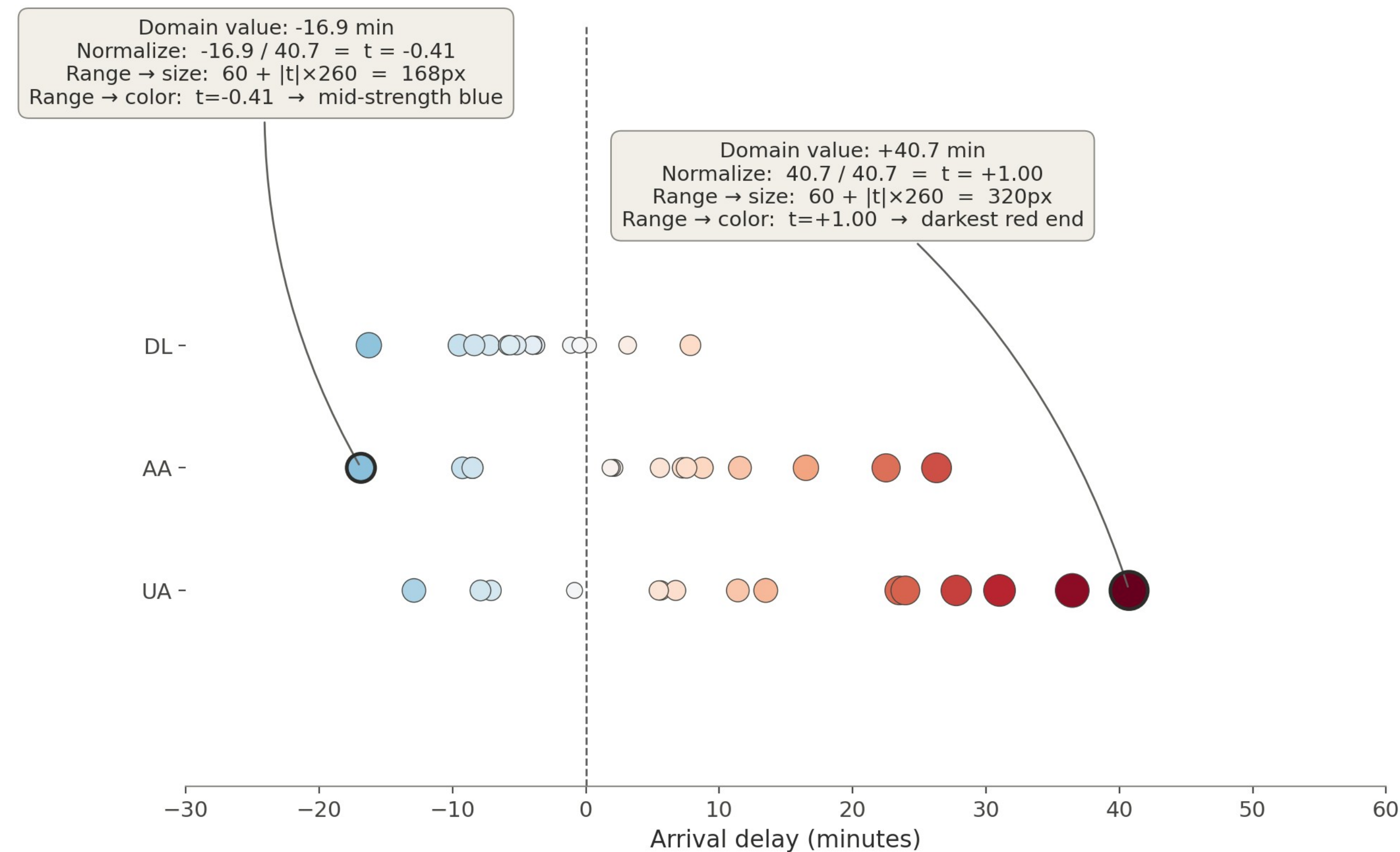
Airlines = [DL, AA, UA]
Arrival delay = [-25,43]



Dimensions of a scale

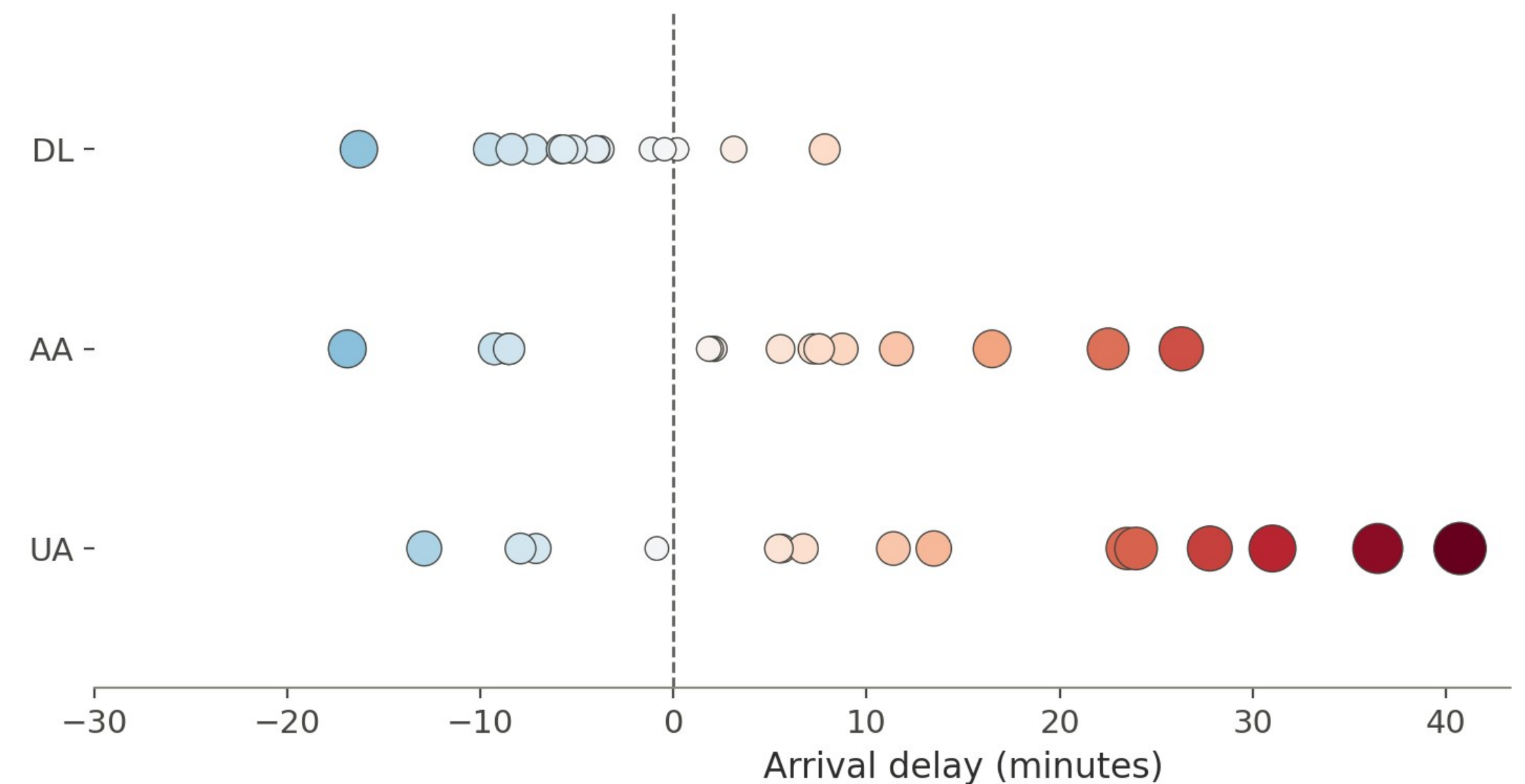
- **Domain:** the data values to be mapped
- **Range:** the visual channel values derived from domain

Airlines = [DL, AA, UA]
Arrival delay = [-30,40]



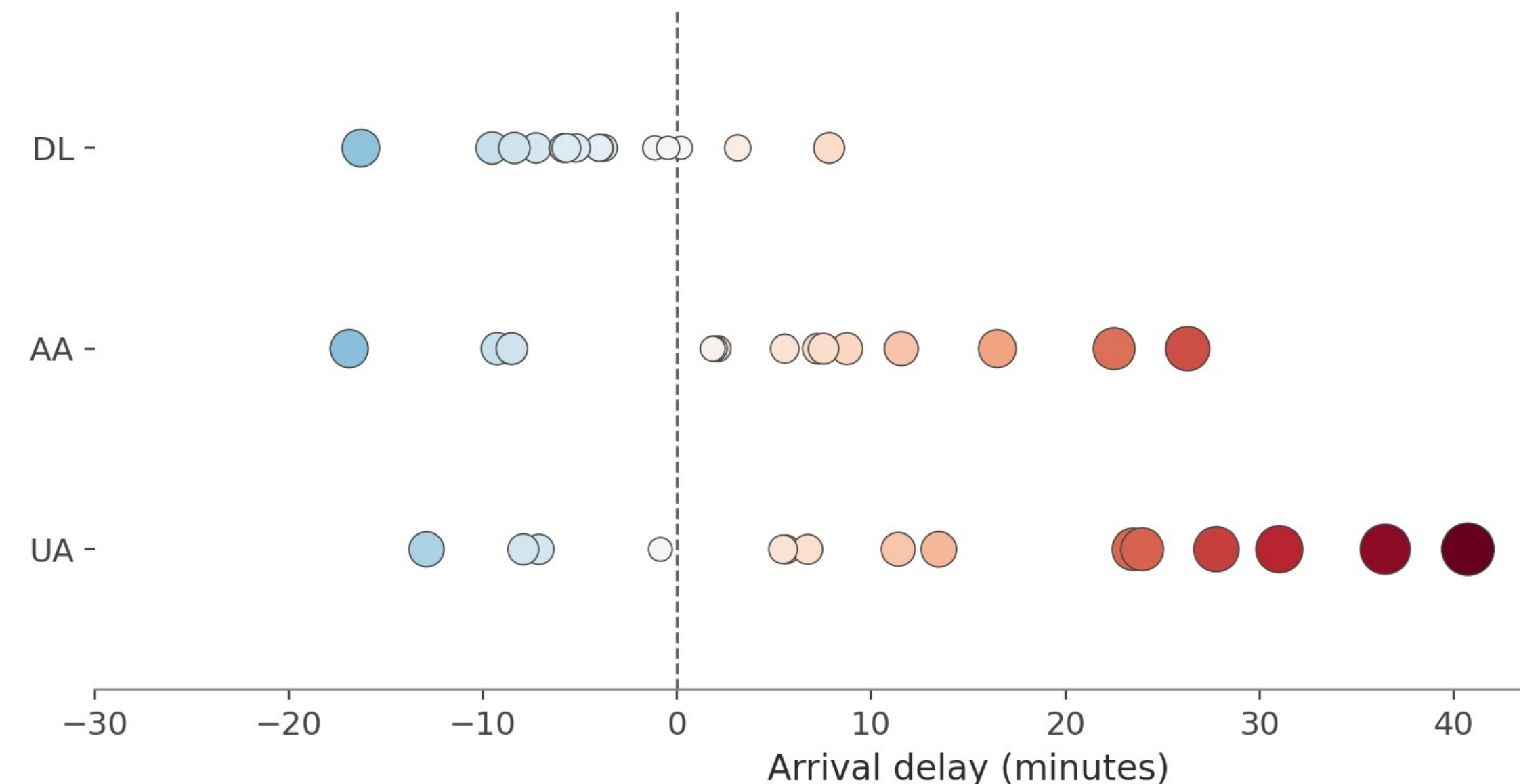
Dimensions of a scale

- **Domain:** the data values to be mapped
- **Range:** the visual channel values derived from domain
- **Type:** sequential, diverging, linear, log, power, square root, ...



Dimensions of a scale

- **Domain:** the data values to be mapped
- **Range:** the visual channel values derived from domain
- **Type:** sequential, diverging, linear, log, power, square root, ...
- **Aggregator:** how values are aggregated if a mark represents multiple items



Decoding charts!

Write out **one mapping**:

- A mapping between **data items and marks**
- A mapping between **data attributes and visual channels**

If all mappings are already written out by other students, write your answer to the following about a particular mapping: **is it a good one?**

For each **ATTRIBUTE → CHANNEL** encoding:

- What is the domain?
- What is the range?
- What is the scale type?
- Is the data being aggregated?